SARCathon 2024 | ConsultQuest

TEAM

CHANAKYA

MUKUND SINGHAL

KALPAK AGARWAL

SANKET BANSAL

VAIBHAV ITAURIYA

PROBLEM STATEMENT

Reimagining Workforce Development in the Era of AI, the Struggle for Quality Education in India's Government Schools

Why this problem need to be solved

Government schools in India are failing to prepare students for the 21st-century workforce due to various challenges, including overburdened teachers, poor infrastructure, outdated curricula and negative public perceptions. As modern jobs increasingly demand critical thinking, problem-solving and technological skills, students from government schools are at a disadvantage, deepening the divide between public and private education systems. This gap risks leaving millions of students unprepared for the evolving job market, threatening their future economic prospects and perpetuating socioeconomic inequality across the nation.

Issues Faced

MAJOR ISSUES	ISSUE DESCRIPTION
OUTDATED CURRICULAM	Failing to meet modern educational needs and technological advancements
INFRASTRUCTURE	Many schools, lack essential educational facilities
ABSENTEEISM	Students and Teachers Absenteeism
SKILLS AND OVERBURDENED	Teachers have to manage large classes and administrative duties with limited training in new teaching methodologies and technologies.

USER PERSONA

Target Audience: Poor Class Families



NAME RANI KUMARI

AGE 13 YEARS

LOCATION OUTSKIRTS OF KANPUR

GRADE 8TH GRADE

GOALS & ASPIRATIONS:	Short-Term Goals: • Perform well in school exams to secure a government scholarship or free tuition • Help her family by tutoring younger children in the neighborhood • Learn English, as it is seen as a path to better opportunities Long-Term Goals: • Become a teacher or government employee to support her family • Break the cycle of poverty by getting a stable job through education
FRUSTRATIONS:	• Lack of Resources: The school lacks basic educational materials, often leaving her without enough textbooks or access to modern learning tools like computers. • Financial Struggles: Her family cannot afford private tuitions or additional educational resources, putting her at a disadvantage compared to better-off peers. • Limited Support for Extracurriculars: There are few opportunities to participate in sports, arts, or other activities due to the lack of school infrastructure and resources.
NEEDS	• Basic Educational Resources: Consistent access to textbooks, notebooks, and stationery. She also needs access to more diverse learning tools like computers and the internet for broader exposure. • Scholarship/Financial Assistance: Rani relies on government scholarships and free midday meals at school to ease the financial burden on her family. • Role Models & Mentors: Needs exposure to successful role models from similar backgrounds who have succeeded through education.

Demographics:

- Background: Low-income family, father works as a daily wage laborer, mother is a homemaker
- Family Income: ₹60,000 – ₹1.2 lakhs per annum (below poverty line)
- School Type: Government school with basic facilities (limited classrooms, shared textbooks, no computer lab, poor infrastructure, limited sports resources)
- Living Conditions: Small, rented one-room home, often without consistent electricity or running water, limited access to the internet

USER PERSONA



NAME AARAV SHARMA

AGE 12 YEARS

LOCATION KANPUR

GRADE 7TH GRADE

Target Audience: Middle Class Families

GOALS & ASPIRATIONS:	Short-Term Goals: • Score well in exams to make parents proud • Be selected for the school's sports team (cricket/badminton) • Learn coding basics (introduced through a school computer class) Long-Term Goals: • Become a doctor/engineer (influenced by family expectations) • Excel in extracurriculars (possibly participate in inter-school competitions)
FRUSTRATIONS:	• Pressure from Parents/Teachers: Constant pressure to perform well academically in a competitive environment, often leading to stress. • Balancing School and Hobbies: Finds it difficult to balance between academics, tuition classes, and personal interests like sports or gaming. • Technology Access vs. Restrictions: While he has access to a smartphone and computer, his screen time is heavily regulated by parents, limiting time spent on gaming or socializing online.
NEEDS	• Educational Support: Additional help to cope with the pressure of competitive exams and assessments (especially as he approaches higher grades). • Creative & Physical Outlets: Enjoys sports and creative activities but often feels they get sidelined due to academic pressures. • Balanced Guidance: Needs emotional and academic support to manage stress while also exploring his interests.

Demographics:

- Background: Middle-class family, parents work in stable jobs (father in IT, mother a school teacher)
- Family Income: ₹8–12 lakhs per annum (typical middle-class range)
- School Type: Private school with decent infrastructure (air-conditioned classrooms, science labs, sports facilities, computer lab, library)
- Living Conditions: Comfortable home, no luxury but essentials covered (3-bedroom apartment, internet, TV, family car)

SUB-PROBLEM PRIORTISATION

Current Challenges

1. Outdated Curriculum: Failing to meet modern educational needs and technological advancements.
2. Overburdened Teachers: Teachers are overwhelmed with managing large classes and administrative duties.
3. Less Skilled Teachers: Limited training opportunities, especially in new teaching methodologies and technologies.
4. No AI Integration: Lack of AI-powered tools to enhance personalized learning and streamline the education process.
5. Large Class Sizes: Minimal interaction between teachers and students due to overcrowded classrooms.

Problem

Outdated Curriculum

Overburdened and Less Skilled Teachers

No AI Integration

Large Class Sizes

Why

These issues are crucial to improving education quality. Addressing them will modernize learning, reduce teacher burden, and enhance student outcomes through personalized learning and better teacher-student interaction.

PROPOSED SOLUTION

Implement AI-driven classrooms, teacher support, chatbots, and modernize the curriculum.

Features

- Smart Classrooms
- Teaching Assistants Program
- Teacher Training Development
- AI-Powered Chatbots
- Curriculum Overhaul

Implementation Plan

STEP 1

Update the curriculum to include **modern courses** like AI, Computer Science (CSE), Data Science, and other in-demand fields. Ensure early exposure to these subjects starting from elementary school.

STEP 2

Implement **smartboards** where AI/smart teachers will conduct live online classes, ensuring high-quality teaching.

STEP 3

Deploy TAs in every classroom to provide **personalized support**, answer student queries, and assist the AI teacher.

STEP 4

Five-level training system for teachers
Levels 1-3: For TAs, focusing on classroom management and personalized interaction.
Levels 4-5: For experienced teachers who will manage student interactions and monitor AI classes.

STEP 5

AI chatbots available in computer labs for students to **clear doubts** independently, promoting self-learning.

SUB-PROBLEM PRIORTISATION

Current Challenges

Government schools in India face serious infrastructural challenges that hinder the quality of education. Many schools lack essential facilities like well-equipped classrooms, libraries, science labs, computer rooms, sports areas and clean sanitation. This limits students' ability to learn effectively and makes it difficult for teachers to deliver lessons. Without these facilities, students are deprived of an environment that promotes modern learning, particularly in areas like technology and problem-solving.

Problem

Lack of adequate infrastructure, hindering effective education

Why

To ensure equal access to quality education

PROPOSED SOLUTION

Hiring Interns from prestigious institutes

Solution

The proposal offers internships for IIT and NIT students to improve government school infrastructure, including libraries, labs, and sports facilities, allowing them to apply their skills for a national cause.

Implementation Plan

STEP 1

Partner with top institutes for internships to recruit students as interns

STEP 2

Identify and prioritize school infrastructure projects

STEP 3

Provide mentorship from engineers and educators

STEP 4

Collaborate with government and educational NGOs

SUB-PROBLEM PRIORTISATION

Current Challenges

There is a widespread belief that government schools offer inferior education compared to private schools. This perception leads many parents to prefer private institutions even when government schools are affordable or free. Despite efforts to improve government schools, this negative image persists and affects enrollment, teacher morale and student confidence.

Problem

Perception of inferior education hinders enrollment

Why

To boost enrollment, improve education and perception

PROPOSED SOLUTION

Promote government school success stories widely

Solution

A public relations campaign should highlight success stories from government schools, showcasing improvements in academics, infrastructure, and student achievements. Sharing these stories on social media can shift public perception, boost confidence in government schools, and drive enrollment growth.

Implementation Plan

STEP 1

Collect and document success stories from gov schools

STEP 2

Social media presence on instagram, facebook etc.

STEP 3

Organising public events and press releases

STEP 4

Organise school open days for local community to see improvement

SUB-PROBLEM PRIORTISATION

Current Challenges

Student Absenteeism: Regular attendance is crucial for quality education, but in many government schools, absenteeism is a major issue.

Teacher Absenteeism: Another critical issue is irregular teacher attendance. Without consistent teaching, student performance and overall school quality suffer. Accountability measures are often weak, allowing this problem to persist.

Problem

Absenteeism in Government Schools

Why

Absenteeism results in learning gaps, which can severely affect performance

PROPOSED SOLUTION

Biometric Fingerprint-Based Attendance System

Solution

A fingerprint-based biometric attendance system for both students and teachers will be implemented. Every subject teacher will bring a portable biometric machine to their classes, to mark their attendance, and then pass the machine to each student to mark their own attendance. This system already exists in most of the top government colleges.

Implementation Plan

STEP 1

Setting up and maintaining the biometric devices.

STEP 2

Syncing data with central servers for real-time tracking.

STEP 3

Providing technical support to teachers for operating the devices.

STEP 4

Attendance data will be automatically sent to a central server (whenever internet is available)

SUB-PROBLEM PRIORTISATION

Current Challenges

(A) Government-provided smartphones and digital devices for students are often misused by their families, leading to interruptions in students' studies.
(B) Many students in government schools lack awareness of higher education opportunities and may not be familiar with prestigious institutions like IITs or other government colleges.
(C) Students do not have sufficient opportunities to participate in extracurricular activities such as sports, arts, music, or debate.

Problem

Misuse of Government-Provided Digital Devices in Schools, lack awareness of higher education opportunities.Lack of Access to Extracurricular Activities in Government Schools

Why

Regular access to study materials is crucial for student performance.
This absence of exposure can lead to diminished aspirations
Extracurricular activities are crucial for the all-around development of students.
They help in building essential life skills like teamwork, leadership, problem-solving, and communication

PROPOSED SOLUTION

Dedicated Tablets for Students, Attendance-Based Incentive System, Dedicated Science Labs

Solution

(A) The tablets will work offline to prevent students from needing continuous internet access, which can be a barrier in rural areas. The devices will come preloaded with:

- Lecture Videos
- Assignments
- Study Materials

(B) A system where students receive incentives based on the student's attendance.

Implementation Plan

STEP 1

The technical assistant will be responsible for loading new content (assignments, etc.) every day or week.

STEP 2

This content will be uploaded via a secure pin that only they can access, ensuring that students cannot tamper with or misuse the device.

STEP 3

Incentives will give an assurity to the parents that their children are properly attending the classes and not skipping the classes.

STEP 4

Involving local scientists, educators, and volunteers to facilitate the workshops, providing students with insights into real-world applications of science.

PROPOSED SOLUTION

Educational Tours to IITs and Nearby Government Colleges, Extracurricular Programs in Government Schools

Solution

Arrange visits for students to various IITs and nearby government colleges. The tours will include guided visits to campuses, interactions with current students and faculty, and exposure to academic and extracurricular activities.
Conduct surveys among students, teachers, and parents to identify interests and the types of extracurricular activities that would be most beneficial and appealing to students.

Implementation Plan

STEP 1

Identify key IITs and government colleges in the region that are willing to host student visits.

STEP 2

Coordinate with the institutions to schedule tours and arrange activities that showcase their programs

STEP 3

Based on the needs assessment, curate a list of extracurricular activities to be offered, prioritizing those that cater to diverse interests and skill levels.

STEP 4

Collaborate with local organizations, NGOs, and community groups to provide additional support
Hire qualified coaches, instructors, and volunteers from local communities

PROPOSED SOLUTIONS

Summary



EdTech Integration and AI-Powered Classrooms



Infrastructure Development through Internships from IITs and NITs



Public Perception Shift through Success Story Campaigns



Biometric Attendance Systems for Reducing Absenteeism



Attendance-Based Incentive System and Syllabus Reforms



Interactive Learning through Vigyaan Bus and Dedicated Science Labs

PITFALLS AND MITIGATIONS

Pitfalls	Mitigations
Resistance from educators, lack of infrastructure in rural areas, over-reliance on AI reducing human interaction and technical disruptions like internet failures.	Offer training, incentives, gradual urban-first rollout with low-cost tech for rural areas, balance AI with human support and build strong backup systems.
Funding constraints, stakeholder coordination issues, interns' lack of practical experience and limited resources for post-implementation maintenance.	Use grants, CSR and fees for funding. Improve coordination with a management team. Train interns with expert oversight. Ensure maintenance through local agreements and regular check-ups.
Gathering authentic rural stories is time-consuming. Social media needs constant updates for engagement and changing negative perceptions of government schools takes persistence.	Partner with schools, use a content calendar for social media, and host community events to boost positive word-of-mouth.
Storing biometric data involves significant risks related to privacy and security. If sensitive data is compromised, it can lead to legal, ethical and reputational issues	Implement strong encryption, restrict access and ensure compliance with privacy regulations to protect biometric data from breaches.
Technical issues in tabs and labs such as device malfunctions and software glitches can disrupt education services and system reliability.	Perform regular maintenance and provide timely technical support to prevent malfunctions and ensure smooth system operation.

VIDEO SUBMISSION

We went to Government School, Nankari (near IIT Kanpur) to know about the situation and level of study there. We talked to the principal and identified key issues.

Then we went to Shiksha Sopan Ashram, IIT Kanpur run by HC Verma Sir which is our ideal model for a government school. This school contained science labs, smart classrooms and was equipped with modern tools.

Thus, this presentation is the conclusion of all our research and analytics.

Video Link: https://www.canva.com/design/DAGUTxRhJXo/BD7lMxZEEaS7LMctxeuv5w/edit?utm_content=DAGUTxRhJXo&utm_campaign=designshare&utm_medium=link2&utm_source=sharebutton

THANK YOU